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GTIIT

CALCULUS 1 - M WINTER SEMESTER - 2024-2025

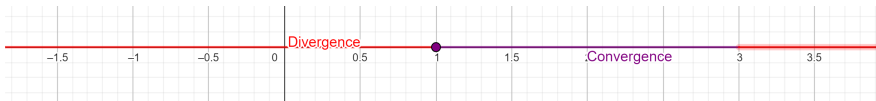
LECTURE 25 - JANUARY 8TH

TAYLOR SERIES: MORE EXAMPLES

POWER SERIES CONVERGENCE THEOREM

If a series $\sum_{n=0}^{\infty} c_n(x - a)^n$ has radius of convergence $R > 0$, then

- 1 The series **converges absolutely** for all $x \in (a - R, a + R)$.
- 2 The series **diverges** for all $x \in (-\infty, a - R) \cup (a + R, \infty)$.



As a corollary

- 3 If $\sum_{n=0}^{\infty} c_n(x - a)^n$ converges for x_0 then it **converges absolutely** for all x such that $|x - a| < |x_0 - a|$

3) Proof

Suppose that $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges for a number x_0 , then

$a - R \leq x_0 \leq a + R$, because the series **diverges for all**
 $x \in (-\infty, a - R) \cup (a + R, \infty)$.

- If $x_0 = a \pm R$, then the conclusion follows from the theorem 1.
 - If $|x_0 - a| < R$, then the conclusion follows from the theorem 1.
- So, the series **converges absolutely for all x such that**
 $|x - a| < |x_0 - a|$ ✓

TABLE. MAIN MACLAURIN SERIES: SEE PREVIOUS LECTURE

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots \quad R = 1$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad R = \infty$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad R = \infty$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \quad R = \infty$$

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \quad R = 1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad R = 1$$

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \dots \quad R = 1$$

Ex 1. MACLAURIN SERIES: COMPUTE THE LIMIT

$$\lim_{x \rightarrow 0} \frac{\sin(x) - x + \frac{1}{6}x^3}{x^5}$$

Solution. • Use L'Hospital's R several times or Maclaurin Series:

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

$$\Rightarrow \frac{\sin(x) - x + \frac{1}{6}x^3}{x^5} = \frac{1}{x^5} \left[\cancel{x} + \cancel{\frac{1}{6}x^3} + \cancel{x - \frac{1}{6}x^3} + \frac{1}{5!}x^5 + \sum_{n=3}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \right]$$

$$= \frac{1}{x^5} \left[\frac{1}{5!}x^5 + \sum_{n=3}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \right] = \frac{1}{5!} + \sum_{n=3}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{\overbrace{2n+1-5}^{\geq 2}}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin(x) - x + \frac{1}{6}x^3}{x^5} = \lim_{x \rightarrow 0} \underbrace{\left[\frac{1}{5!} + \sum_{n=3}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1-5} \right]}_{\rightarrow 0} = \frac{1}{5!} = \frac{1}{120} \checkmark$$

MACLAURIN SERIES

Exercise 2. See Ex. 16 of Lecture 24. Then, find the Maclaurin series for

a) $f(x) = e^{-x^2}$

b) $F(x) = \int e^{-x^2} dx$

c) Compute $\int_0^1 e^{-x^2} dx$ with accuracy of 10^{-3} .

Maclaurin Series 2. a)

- The Theorem implies that $g(t) = e^t$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$e^t = \sum_{n=0}^{\infty} \frac{t^n}{n!} \text{ for all } x \in \mathbb{R}$$

- Use the Taylor series for $g(t) = e^t$ and compose with $t = -x^2 \forall t \in \mathbb{R}$:

$$e^{-x^2} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!} \text{ for all } x \in \mathbb{R} \checkmark$$

Antiderivative of the previous function

b) Taylor Theorem allows us to integrate on the interval of convergence, which is all $\mathbb{R}$, to find the Taylor series of the antiderivative:

$$\int e^{-x^2} dx = \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \int x^{2n} dx = C + \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \frac{1}{(2n+1)} x^{2n+1} \quad \forall x \in \mathbb{R} \quad \checkmark$$

Ex 2. c) Apply the Fundamental Theorem of Calculus

$$\int_0^1 e^{-x^2} dx = \left[C + \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \frac{1}{(2n+1)} x^{2n+1} \right] \Big|_{x=0}^{x=1} = \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \frac{1}{(2n+1)}$$

$$= 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} - \dots$$

We know by the Alternating Series Estimation Theorem

$$0 \leq |R_n| \leq |a_{n+1}| \leq \frac{1}{(n+1)!(2n+3)} < 10^{-3} \text{ for } n \geq 4$$

because $n = 4 \implies 5!(2 \cdot 4 + 3) = 120 \cdot 11 > 1200 > 10^3$ ✓

Hence, let us approximate the definite integral with the 5 first terms of the series, namely, the sum from $n = 0$ to $n = 4$:

$$\int_0^1 e^{-x^2} dx \approx 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} \approx 0.7475 \checkmark$$

True Value: $\int_0^1 e^{-x^2} dx \approx 0.7468 \implies |R_5| = |0.7475 - 0.7468| \approx 0.0007 < 0.001 \checkmark$

PASCAL'S TRIANGLE: BINOMIAL COEFFICIENTS

Define the **Combinatorial Numbers**, for $k, n \in \mathbb{N}$ or $n = 0$:

$$\binom{k}{0} = 1, \quad \binom{k}{n} = \frac{\overbrace{k(k-1)(k-2) \cdots (k-n+1)}^{n \text{ factors}}}{n!} = \frac{k!}{n!(k-n)!}$$

$$\binom{0}{0}$$

$$\binom{1}{0} \binom{1}{1}$$

$$\binom{2}{0} \binom{2}{1} \binom{2}{2}$$

$$\binom{3}{0} \binom{3}{1} \binom{3}{2} \binom{3}{3}$$

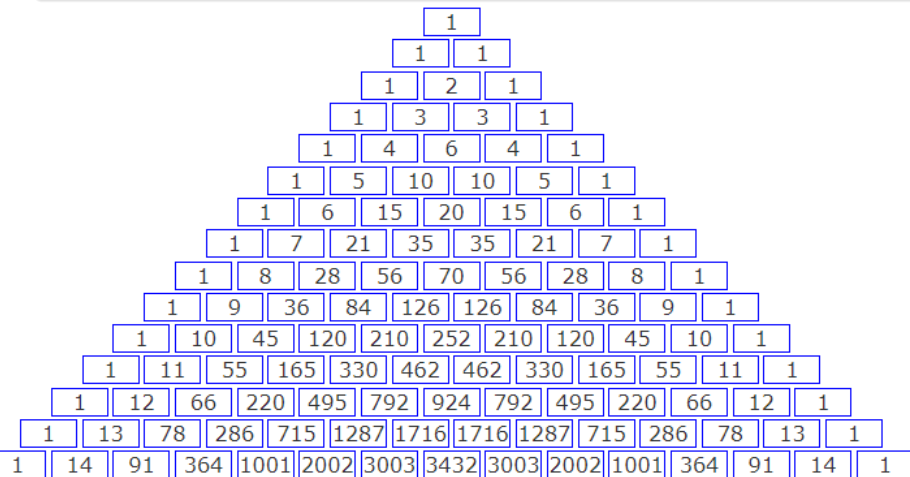
$$\binom{4}{0} \binom{4}{1} \binom{4}{2} \binom{4}{3} \binom{4}{4}$$

$$\binom{5}{0} \binom{5}{1} \binom{5}{2} \binom{5}{3} \binom{5}{4} \binom{5}{5}$$

$$\binom{6}{0} \binom{6}{1} \binom{6}{2} \binom{6}{3} \binom{6}{4} \binom{6}{5} \binom{6}{6}$$

$$\binom{7}{0} \binom{7}{1} \binom{7}{2} \binom{7}{3} \binom{7}{4} \binom{7}{5} \binom{7}{6} \binom{7}{7}$$

BINOMIAL COEFFICIENTS AND BINOMIAL SERIES.



The row $k + 1$ contains the $k + 1$ combinatorial numbers $\binom{k}{n} : 0 \leq n \leq k$

BINOMIAL COEFFICIENTS AND BINOMIAL SERIES. EXAMPLES IN THE FINITE CASE

- $(1+x)^2 = 1 + 2x + x^2 = \binom{2}{0} + \binom{2}{1}x + \binom{2}{2}x^2$
- $(1+x)^3 = 1 + 3x + 3x^2 + x^3 = \binom{3}{0} + \binom{3}{1}x + \binom{3}{2}x^2 + \binom{3}{3}x^3$
- $(1+x)^4 = (1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$
$$= \binom{4}{0} + \binom{4}{1}x + \binom{4}{2}x^2 + \binom{4}{3}x^3 + \binom{4}{4}x^4$$
- $(1+x)^5 = (1+x)^5 = 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$
$$= \binom{5}{0} + \binom{5}{1}x + \binom{5}{2}x^2 + \binom{5}{3}x^3 + \binom{5}{4}x^4 + \binom{5}{5}x^5$$

BINOMIAL SERIES. DEFINITION AND THEOREM.

- 1 Define the binomial coefficients for any $k \in \mathbb{R}$ and $n \in \mathbb{N}$ or $n = 0$:

$$\binom{k}{n} = \frac{\overbrace{k(k-1)(k-2) \cdots (k-n+1)}^{n \text{ factors}}}{n!}$$

- 2 If k is any real number and $|x| < 1$, then the Maclaurin Series of the binomial function follows

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n$$

- 3 The interval of convergence is $(-1, 1]$ if $-1 < k < 0$.
- 4 The interval of convergence is $[-1, 1]$ if $k \geq 0$.
- 5 If $k \in \mathbb{N}$, then $\binom{k}{n} = 0 \forall n \geq k$, and the series is the finite sum $(1+x)^k = \sum_{n=0}^k \binom{k}{n} x^n$

PROOF OF THE BINOMIAL SERIES.

- Compute the Maclaurin coefficients $c_n = \frac{f^{(n)}(0)}{n!}$

$$f(x) = (1+x)^k \implies f(0) = 1$$

$$f'(x) = k(1+x)^{k-1} \implies f'(0) = k$$

$$f''(x) = k(k-1)(1+x)^{k-2} \implies f''(0) = k(k-1)$$

$$f^{(3)}(x) = k(k-1)(k-2)(1+x)^{k-3} \implies f^{(3)}(0) = k(k-1)(k-2)$$

... ..

$$f^{(n)}(x) = k(k-1)\dots(k-n+1)(1+x)^{k-n} \implies f^{(n)}(0) = k(k-1)\dots(k-n+1)$$

Then, the Maclaurin series is

$$\implies (1+x)^k = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{k(k-1)\dots(k-n+1)}{n!} x^n = \sum_{n=0}^{\infty} \binom{k}{n} x^n$$

CONVERGENCE OF THE BINOMIAL SERIES

- If $k \in \mathbb{N} \implies \binom{k}{k+1} = 0$, since it includes the factor $(k - k) = 0$ and this factor is in all terms for $n > k \implies$ the series is a **finite sum**:

$$(1+x)^k = \sum_{n=0}^k \binom{k}{n} x^n$$

- For any $k \notin \mathbb{N}$, apply the Ratio Test:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{k(k-1)\dots(k-n+1)(k-n)|x|^{n+1}}{(n+1)!} \cdot \frac{n!}{k(k-1)\dots(k-n+1)|x|^n} = \frac{|k-n|}{(n+1)} |x| = \frac{\left|1 - \frac{k}{n}\right|}{1 + \frac{1}{n}} |x|$$

$$\implies \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{\left|1 - \frac{k}{n}\right|}{1 + \frac{1}{n}} |x| = |x|, \text{ then}$$

- The series converges if $|x| < 1$
- The series diverges if $|x| > 1$ ✓

We've obtained $R = 1$, then the series converges to $f(x)$ at any $x \in (-1, 1)$.

The convergence at the endpoints depends on k and requires further proofs:

- If $k > 0$ and $x = -1 \implies (1+x)^k = 0 = \sum_{n=0}^{\infty} \binom{k}{n} (-1)^n$ ✓
- If $k > 0$ and $x = 1 \implies (1+x)^k = 2^k = \sum_{n=0}^{\infty} \binom{k}{n}$ ✓.
- If $k < 0$ then $(1+x)^k = \frac{1}{(1+x)^{-k}}$ is not defined at $x = -1$.
- If $k < 0$ and $x = 1$ then $(1+x)^k = \frac{1}{2^{-k}} = \sum_{n=0}^{\infty} \binom{k}{n}$.

BINOMIAL SERIES.

3. Example. Find the Maclaurin series of $f(x) = \frac{1}{\sqrt{4-x}}$

Solution. $k = -\frac{1}{2}$, $\binom{k}{n} = \frac{\overbrace{k(k-1)(k-2)\cdots(k-n+1)}^{n \text{ factors}}}{n!}$.

$$\begin{aligned}\frac{1}{\sqrt{4-x}} &= \frac{1}{2} \left(1 - \frac{x}{4}\right)^{-1/2} = \frac{1}{2} \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \left(-\frac{x}{4}\right)^n \\&= \frac{1}{2} \left[1 + \binom{-\frac{1}{2}}{1} \left(-\frac{x}{4}\right) + \frac{\binom{-\frac{1}{2}}{2} \left(-\frac{x}{4}\right)^2}{2!} + \frac{\binom{-\frac{1}{2}}{3} \left(-\frac{x}{4}\right)^3}{3!} \right. \\&\quad \left. + \cdots + \frac{\binom{-\frac{1}{2}}{n} \left(-\frac{x}{4}\right)^n}{n!} + \cdots \right] \\&= \frac{1}{2} \left[1 + \frac{1}{8}x + \frac{1 \cdot 3}{2! \cdot 8^2}x^2 + \frac{1 \cdot 3 \cdot 5}{3! \cdot 8^3}x^3 + \cdots + \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n! \cdot 8^n}x^n + \cdots \right]\end{aligned}$$

Hence, for all $x \in (-4, 4)$:

$$f(x) = \frac{1}{\sqrt{4-x}} = \frac{1}{2} \cdot \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \frac{(-1)^n}{4^n} x^n :$$

$$= \frac{1}{2} \cdot \sum_{n=0}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n! 8^n} x^n \quad \checkmark$$

Ex. 4. Given $f(x) = \int_0^x \frac{\sin(t)}{(\pi - t)} dt$, find $\int_0^\pi f(x) dx$.

Solution.

① Taylor Series of $\sin(x)$ at $a = \pi$, convergent $\forall x \in \mathbb{R}$:

- $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$

- Use the identity and compose with $x = t - \pi$:

$$\sin(t) = \sin(t - \pi) \underbrace{\cos(\pi)}_{=-1} + \cos(t - \pi) \underbrace{\sin(\pi)}_{=0} = (-1) \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (t - \pi)^{2n+1}$$

- Divide by $-(t - \pi)$

$$\Rightarrow \frac{\sin(t)}{(\pi - t)} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (t - \pi)^{2n} \quad \forall t \in \mathbb{R} \checkmark$$

2 Integrate $f(x) = \int_0^x \frac{\sin(t)}{(\pi - t)} dt = \int_0^x \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (t - \pi)^{2n} \right] dt$

$$= \sum_{n=0}^{\infty} \left[\frac{(-1)^n}{(2n+1)!} \int_0^x (t - \pi)^{2n} dt \right] = \sum_{n=0}^{\infty} \left[\frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+1)} (t - \pi)^{2n+1} \right] \Bigg|_{t=0}^{t=x}$$

$$= \sum_{n=0}^{\infty} \left[\frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+1)} (x - \pi)^{2n+1} \right] + \sum_{n=0}^{\infty} \left[\frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+1)} \pi^{2n+1} \right]$$

3

Compute the definite integral:

$$\int_0^{\pi} f(x) dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+1)(2n+2)} (x - \pi)^{2n+2} \Big|_{x=0}^{x=\pi} + \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+1)} \pi^{2n+2}$$

$$= - \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+1)(2n+2)} \pi^{2n+2} + \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+1)} \pi^{2n+2}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+1)} \pi^{2n+2} \underbrace{\left[\frac{-1}{(2n+2)} + 1 \right]}_{\frac{(2n+1)}{(2n+2)}}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+2)!} \pi^{2n+2} \leftarrow k = n + 1$$

$$= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(2k)!} \pi^{2k} = - \left[\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \pi^{2k} - 1 \right] = - [\cos(\pi) - 1]$$

So, the definite integral equals $\int_0^{\pi} f(x) dx = +2 \checkmark$

Ex 5. Answer each of the following True or False questions:

a) If $\sum_{n=0}^{\infty} (-1)^n a_n$ converges and $\sum_{n=0}^{\infty} a_n$ doesn't converge, then the interval of convergence of $\sum_{n=0}^{\infty} a_n (x - 2)^n$ is $[1, 3)$.

True

Solution. • Let R be the radius of convergence, so, the series $\sum a_n (x - 2)^n$ diverges for all $x \in (-\infty, 2 - R) \cup (2 + R, \infty)$.



• $x = 1$ belongs to the interval of convergence

$$\implies 2 - R \leq 1 \leq 2 + R \implies 2 - 1 \leq R \implies 1 \leq R$$

FINAL EXAM A - SPRING 2023

- But $x = 3$ doesn't belong to the interval of convergence

$$\implies 2 + R \leq 3 \implies R \leq 1 \implies R = 1$$

- $|x - 2| = 1 \iff x = 1 \text{ or } x = 3$

- Hence, the interval of convergence is $[1, 3)$ ✓

FINAL EXAM A - SPRING 2023. EX 5. ANSWER EACH OF THE FOLLOWING TRUE OR FALSE QUESTIONS:

c) If $\sum_{n=0}^{\infty} n^2 \cdot a_n$ converges, then $\sum_{n=0}^{\infty} a_n$ converges absolutely.

True

Solution. • Since $\sum_{n=0}^{\infty} n^2 \cdot a_n$ converges $\implies \lim_{n \rightarrow \infty} n^2 \cdot a_n = 0$.

• Given $\epsilon = 1 \exists N : n > N \implies |n^2 \cdot a_n| < 1$

• Take $M = \text{Max}\{1, |n^2 \cdot a_n| : 1 \leq n \leq N\}$

$$\implies |n^2 \cdot a_n| \leq M \forall n \implies 0 \leq |a_n| \leq \frac{M}{n^2} \forall n$$

• The series $\sum_{n=0}^{\infty} \frac{M}{n^2}$ converges $\implies \sum_{n=0}^{\infty} |a_n|$ converges ✓

Ex 6) Find a rational number whose distance from $\sqrt{4.5}$ is at most $\frac{1}{2000} = \frac{5}{10^4}$.

- Compute the first coefficients $c_n = \frac{f^{(n)}(4)}{n!}$ of the Taylor series of $f(x) = \sqrt{x}$ at $a = 4$:

$$f(x) = x^{1/2} \implies f(4) = 2$$

$$f'(x) = \frac{1}{2}x^{-1/2} \implies f'(4) = \frac{1}{4}$$

$$f''(x) = -\frac{1}{4}x^{-3/2} \implies f''(4) = -\frac{1}{32}$$

$$f^{(3)}(x) = \frac{3}{8}x^{-5/2} \implies f^{(3)}(4) = \frac{3}{256}$$

$$f^{(4)}(x) = -\frac{15}{16}x^{-7/2} \implies f^{(4)}(4) = -\frac{15}{2048}$$

$$f^{(5)}(x) = -\frac{15 \cdot 7}{32}x^{-9/2} \implies f^{(5)}(4) = -\frac{105}{16384}$$

In general, compose with the Binomial Formula:

$$(1 + u)^k = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} u^n = \sum_{n=0}^{\infty} \frac{k(k-1)\dots(k-n+1)}{n!} u^n$$

with $k = \frac{1}{2}$, $u = \frac{(x-4)}{4}$:

$$\Rightarrow f(x) = \sqrt{x} = 2\sqrt{1 + \frac{(x-4)}{4}} = 2 \left[\sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \frac{1}{4^n} (x-4)^n \right]$$

$$\begin{aligned} &= 2 \left[1 + \binom{\frac{1}{2}}{1} \frac{1}{4^1} (x-4) + \binom{\frac{1}{2}}{2} \frac{1}{4^2} (x-4)^2 + \binom{\frac{1}{2}}{3} \frac{1}{4^3} (x-4)^3 + \binom{\frac{1}{2}}{4} \frac{1}{4^4} (x-4)^4 + \dots \right] \\ &= 2 \left[1 + \frac{1}{2} \frac{1}{4} (x-4) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!} \frac{1}{4^2} (x-4)^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!} \frac{1}{4^3} (x-4)^3 \right. \\ &\quad \left. + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{4!} \frac{1}{4^4} (x-4)^4 + \dots \right] \end{aligned}$$

Taylor series of $f(x) = \sqrt{x}$ at $a = 4$:

$$\sqrt{x} = 2 + \frac{1}{4}(x - 4) - \frac{1}{64}(x - 4)^2 + \frac{1}{512}(x - 4)^3 - \frac{5}{16384}(x - 4)^4 + \dots$$

Recall that the Taylor polynomial of degree n approximates the function at $x = 4.5$ with a remainder:

$$f(4.5) = T_n(4.5) + R_n(4.5)$$

For instance, if $n = 3$:

$$\sqrt{4.5} = f(4.5) = T_3(4.5) + R_3(4.5)$$

So, if we take off the term of the reminder, we obtain an approximation:

$$\sqrt{4.5} \approx \left[2 + \frac{1}{4}(x - 4) - \frac{1}{64}(x - 4)^2 + \frac{1}{512}(x - 4)^3 \right] \Big|_{x=4.5}$$

$$\sqrt{4.5} \approx \left[2 + \frac{1}{4}\left(\frac{1}{2}\right) - \frac{1}{64}\left(\frac{1}{2}\right)^2 + \frac{1}{512}\left(\frac{1}{2}\right)^3 \right] = 2.121337$$

- The numerical series of the previous page is an alternating series, so the remainder is bounded by the absolute value of the next coefficient, then

$$|R_n| \leq |a_{n+1}| \text{ and } R_3 \leq \frac{5}{16384} < \frac{5}{10^4} \implies n \geq 3$$

$$\implies \sqrt{4.5} \approx T_3[f(4.5)] \approx \left[2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2 + \frac{1}{512}(x-4)^3 \right] \Big|_{x=4.5}$$

$$\approx \left[2 + \frac{1}{4}\left(\frac{1}{2}\right) - \frac{1}{64}\left(\frac{1}{2}\right)^2 + \frac{1}{512}\left(\frac{1}{2}\right)^3 \right] = 2.121337$$

$$R_3 = |\sqrt{4.5} - 2.12134| \approx |2.121320 - 2.121337| = 0.000017 < 10^{-4} \checkmark$$

Ex. 7. Let $g(x) = (1 + \frac{\pi}{4})x - \frac{1}{2}x^2$ and $h(x) = x \arctan(x) - \frac{1}{2} \ln(x^2 + 1)$. Find the local and absolute extreme values on $(0, +\infty)$ of

$$f(x) = \int_0^{g(x)-h(x)} \left(\frac{t^4 + 3}{e^t + 1} \right) dt.$$

Solution. • Because of the Fundamental Theorem of Calculus, f is differentiable and

$$f'(x) = \overbrace{\left(\frac{t^4 + 3}{e^t + 1} \right)}^{>0} \Big|_{t=g(x)-h(x)} (g'(x) - h'(x))$$

- Compute the derivatives: $g'(x) - h'(x) = 1 + \frac{\pi}{4} - x - \arctan(x)$
- Since the integrand is > 0 for all $t \in \mathbb{R}$, the roots of f' are only the roots of $g'(x) - h'(x) = 1 + \frac{\pi}{4} - x - \arctan(x)$

- One root is easy to find, this is $f'(1) = 0$, because $\arctan(1) = \frac{\pi}{4}$

$$\implies \left[g'(x) - h'(x) \right] \Big|_{x=1} = \left[1 + \frac{\pi}{4} - x - \arctan(x) \right] \Big|_{x=1} = 1 + \frac{\pi}{4} - 1 - \frac{\pi}{4} = 0$$

- $\frac{t^4+3}{e^t+1} > 0 \ \forall t$ then $f' > 0$ if and only if $g'(x) - h'(x) > 0$:

$\arctan$ is increasing, so, for

$$0 < x < 1 \implies \arctan(x) < \arctan(1) = \frac{\pi}{4}$$

$$\implies 1 + \frac{\pi}{4} - x - \arctan(x) > 1 + \frac{\pi}{4} - 1 - \frac{\pi}{4} \geq 0$$

So,

$$f' > 0 \text{ for } x \in (0, 1).$$

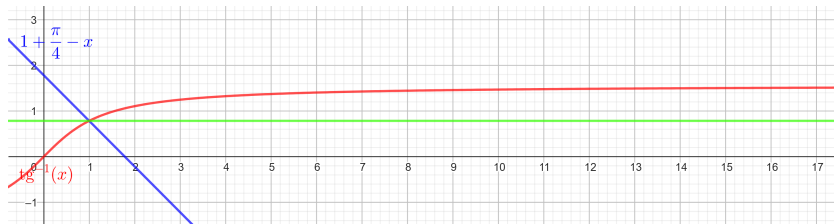
- $\frac{t^4+3}{e^t+1} > 0 \forall t$ then $f' < 0$ if and only if $g'(x) - h'(x) < 0$:

$$\text{For } 1 < x \implies \arctan(1) = \frac{\pi}{4} < \arctan(x)$$

$$\implies 1 + \frac{\pi}{4} - x - \arctan(x) < 1 + \frac{\pi}{4} - 1 - \frac{\pi}{4} \leq 0$$

So,

$$f' < 0 \text{ for } x \in (1, \infty).$$



Hence

	$(0, 1)$	1	$(1, \infty)$
$g'(x) - h'(x)$	> 0	$= 0$	< 0
f'	> 0	$= 0$	< 0
f	<i>increases</i>	loc Max	<i>decreases</i>

We conclude that f has a local and absolute maximum at $x = 1$ with value $M = f(1)$ ✓.